

Aimin Li  *Postdoctoral Researcher, Middle East Technical University*

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🌐 Homepage: [aiminli-hi.github.io](https://github.com/aiminli-hi) · 🔗 LinkedIn: [LinkedIn Link](#)
🎓 Google Scholar: [Google Link](#) · 📍 Address: Ankara, Turkey

RESEARCH PROFILE

Research Identity Information freshness for future networked intelligence: age-optimal control, goal-oriented communications, and intermittent satellite/space networks.

Current Agenda Freshness-aware learning and control for space-air-ground systems, autonomous mobility, energy-constrained sensing, and secure/quantum-era communications.

EDUCATION

Harbin Institute of Technology, Shenzhen (985 Project, C9 League) 2020–2025
Ph.D. in Electrical Engineering ADVISOR: PROF. SHAOHUA WU
Honors: **Best Dissertation Award Nomination** (one of only 2 nominees in the School); Special Merit Doctoral Scholarship; National Scholarship of China (*twice*); Provincial Outstanding Graduate Award.

Harbin Institute of Technology, Shenzhen (HITSZ) (985 Project, C9 League) 2016–2020
B.S. in Electrical Engineering ADVISOR: PROF. SHAOHUA WU
Honors: **Best Thesis Award**; Provincial Merit Student; First Prize of the Chinese Mathematics Competitions (*twice*).

RESEARCH EXPERIENCE

Middle East Technical University (METU), Ankara, Turkey 2025–Present
Postdoctoral Researcher ADVISOR: PROF. ELIF UYSAL
Researcher on an ERC Advanced Grant project; information freshness, age-optimal control, status-update systems, and intermittent communication networks.

Agency for Science, Technology, and Research (A*STAR), Singapore 2023–2024
Visiting Researcher, Institute for Infocomm Research (I²R) ADVISOR: PROF. SUMEI SUN
Research on finite-blocklength coding, goal-oriented communications, and semantic communications.

FULL PUBLICATIONS¹

Underlined first authors are student/junior collaborators I mentored; **(M)** marks a mentored student-first-author outcome.

A. First-Author Publications

Journal Articles

1.  **Aimin Li**, Shaohua Wu^{*}, Gary Lee, and Sumei Sun. “From Freshness to Effectiveness: Goal-oriented Sampling for Remote Decision Making”, *IEEE Trans. Inf. Theory*. (IF: 2.9, JCR Q1) 2025
2.  **Aimin Li**, Xiaomeng Chen, Shaohua Wu^{*}, Gary C.F. Lee, and Sumei Sun. “Unified Upper Bounds on the ML Decoding Error Probability of Spinal Codes over Fading Channels”, *IEEE Trans. Wirel. Commun.* (IF: 10.7, JCR Q1) 2025
3.  **Aimin Li**, Shaohua Wu^{*}, Xiaomeng Chen, and Sumei Sun. “Error Floor of ML-Decoded Spinal Codes in the Finite Blocklength Regime”, *IEEE Trans. Veh. Technol.* (IF: 7.1, JCR Q1) 2025
4.  **Aimin Li**, Shaohua Wu^{*}, Xiaomeng Chen, and Sumei Sun. “Tight Upper Bounds on the BLER of Spinal Codes over the AWGN Channel”, *IEEE Trans. Commun.*, vol. 73, no. 1, pp. 332–347. (IF: 8.9, JCR Q1) 2024
5.  **Aimin Li**, Shaohua Wu^{*}, Siqi Meng, Rongxing Lu, Sumei Sun, and Qinyu Zhang. “Toward Goal-Oriented Semantic Communications: New Metrics, Framework, and Open Challenges”, *IEEE Wirel. Commun.*, vol. 31, no. 5, pp. 238–245. (IF: 11.5, JCR Q1) 2024
6.  **Aimin Li**, Shaohua Wu^{*}, Sumei Sun, and Jie Cao. “Goal-Oriented Tensor: Beyond Age of Information Towards Semantics-Empowered Goal-Oriented Communications”, *IEEE Trans. Commun.*, vol. 72, no. 12, pp. 7689–7704. (IF: 8.9, JCR Q1) 2024
7.  **Aimin Li**, Shaohua Wu^{*}, Jian Jiao, Ning Zhang, and Qinyu Zhang. “Age of Information with Hybrid-ARQ: A Unified Explicit Result”, *IEEE Trans. Commun.*, vol. 70, no. 12, pp. 7899–7914. (IF: 8.9, JCR Q1) 2022
8.  **Aimin Li**, Shaohua Wu^{*}, Jian Jiao, Ning Zhang, and Qinyu Zhang. “Spinal Codes over Fading Channel: Error Probability Analysis and Encoding Structure Improvement”, *IEEE Trans. Wirel. Commun.*, vol. 20, no. 12, pp. 8288–8300. (IF: 10.7, JCR Q1) 2021

Conference Proceedings

¹(★ indicates my advisor, # indicates co-first authors, and † indicates corresponding author.)

9. **[ISIT]** **Aimin Li**, Yigit Ince, and Elif Uysal*. "Taming the Heavy Tail: Age Optimal Preemption",
IEEE International Symposium on Information Theory (IEEE ISIT) 2026
10. **[ALLERTON]** **Aimin Li**, and Elif Uysal*. "Optimal Sampling and Scheduling for Remote Fusion Estimation of Correlated Wiener Processes",
61st Allerton Conference on Communication, Control, and Computing (Allerton) 2025
11. **[ITW]** **Aimin Li**, Shaohua Wu*, Gary C.F. Lee, Xiaomeng Chen, and Sumei Sun. "Sampling to Achieve the Goal: An Age-aware Remote Markov Decision Process",
IEEE Information Theory Workshop (IEEE ITW; Session Chair of AoI Session) 2024
12. **[WCNC]** **Aimin Li**, Shaohua Wu*, and Sumei Sun. "Goal-oriented Tensor: Beyond AoI Towards Semantics-Empowered Goal-oriented Communications",
IEEE Wireless Communications and Networking Conference (IEEE WCNC) 2024
13. **[GLOBECOM]** **Aimin Li**, Shaohua Wu*, Yajing Deng, Jian Jiao, Ning Zhang, and Qinyu Zhang. "Analyzing Age Performance of Hybrid-ARQ: A Unified Explicit Result",
IEEE Global Communications Conference (IEEE GLOBECOM) 2022
14. **[VTC]** **Aimin Li**, Shaohua Wu*, Ying Wang, Jian Jiao, and Qinyu Zhang. "Spinal Codes over BSC: Error Probability Analysis and the Puncturing Design",
IEEE Vehicular Technology Conference (VTC2020-Spring) 2020
15. **[ISIT]** Xiaomeng Chen[#], **Aimin Li**[#], and Shaohua Wu*. "Tight Upper Bounds on the Error Probability of Spinal Codes over Fading Channels",
IEEE International Symposium on Information Theory (IEEE ISIT) (M) 2023

B. Collaborative Publications

Journal Articles

16. **[TMC]** Haoxu Wu, Shaohua Wu*, **Aimin Li**, Siqi Meng, and Qinyu Zhang. "AoI-Aware Joint Sampling-Buffering-Routing Optimization for Autonomous UAV Swarms via a MARL Approach",
IEEE Trans. Mob. Comput. (Accepted) (M) 2026
17. **[TWC]** Shaohua Wu*, Zhihong Deng, **Aimin Li**, Jian Jiao, Ning Zhang, and Qinyu Zhang. "Minimizing Age-of-Information in HARQ-CC Aided NOMA Systems",
IEEE Trans. Wirel. Commun., vol. 22, no. 2, pp. 1072-1086. (IF: 8.9, JCR Q1) 2022
18. **[TCOM]** Siqi Meng, Shaohua Wu*, **Aimin Li**, Jian Jiao, Ning Zhang, and Qinyu Zhang. "Analysis and Optimization of HARQ Based Spinal Coded Timely Status Update System",
IEEE Trans. Commun., vol. 70, no. 10, pp. 6425-6440. (IF: 8.9, JCR Q1) (M) 2022
19. **[TCOM]** Mingxiao Sun, Lulu Song, Di Zhang, Shaobo Jia, **Aimin Li**, Al-Dulaimi, and Shahid Mumtaz. "Information Freshness and Timeliness Analysis in the Finite Blocklength Regime for Mission-Critical Applications",
IEEE Trans. Commun. (IF: 8.9, JCR Q1) (M) 2025
20. **[IOTJ]** Xiaomeng Chen, Shaohua Wu*, **Aimin Li**, Dawei Chen, Pengfei Duan, and Qinyu Zhang. "Beyond Freshness, Towards Uncertainty: Entropy-Driven Sampling for Remote Estimation in Delay-Tolerant Networks",
IEEE Internet of Things J. (IF: 8.9, JCR Q1) (M) 2025
21. **[TVT]** Siqi Meng, Shaohua Wu*, **Aimin Li**, Jian Jiao, Ning Zhang, and Qinyu Zhang. "Partial Self-Concatenation Structure and Performance Analysis of Spinal Codes over Rayleigh Fading Channel",
IEEE Trans. Veh. Technol., vol. 71, no. 6, pp. 6767-6771. (IF: 7.1, JCR Q1) (M) 2022
22. **[NET]** Siqi Meng, Shaohua Wu*, Hanyu Wu, **Aimin Li**, and Qinyu Zhang. "Effectiveness-Oriented SAGSIN: Unveiling a Unified Metric and a Comprehensive Framework",
IEEE Netw., vol. 38, no. 6, pp. 39-47. (IF: 6.3, JCR Q1) (M) 2024
23. **[OJCOM]** Jiahong Ning, **Aimin Li**, Gary C.F. Lee, Tingting Yang, and Sumei Sun. "MARHLO: Hybrid Task Offloading in Maritime MEC via Multi-Agent Reinforcement Learning",
IEEE Open J. Commun. Soc., vol. 11, pp. 10322-10337 (IF: 6.1, JCR Q1) 2025
24. **[JSAIT]** Siqi Meng, Shaohua Wu*, **Aimin Li**, and Qinyu Zhang. "Towards Goal-Oriented Semantic Communications: AoII Analysis of Coded Status Update System Under FBL Regime",
IEEE J. Sel. Areas Inf. Theory, vol. 4, pp. 2641-8770. (M) 2023
25. **[LNET]** Xiyu Lv, Shaohua Wu*, **Aimin Li**, Jian Jiao, Ning Zhang, and Qinyu Zhang. "A Weighted Graph-based Handover Strategy for Aeronautical Traffic in LEO SatCom Networks",
IEEE Netw. Lett., vol. 4, no. 3, pp. 132-136. 2022

Conference Proceedings

26. **[MOBIHOC]** Adem Utku Atasayar, **Aimin Li**, Cagri Ari, and Elif Uysal*. "Fresh Data Delivery: Joint Sampling and Routing for Minimizing the Age of Information?",
ACM MobiHoc (Acceptance Rate: 23%; Only 39 papers accepted worldwide) (M) 2025
27. **[ICC]** Kunpeng Liu, Shaohua Wu*, **Aimin Li**, and Qinyu Zhang. "Joint Transmission and Control in a Goal-oriented NOMA Network",
IEEE International Conference on Communications (IEEE ICC) (M) 2024

28. [ITW](#) Xiaomeng Chen, **Aimin Li**, and Shaohua Wu*. "Optimal Sampling for Uncertainty-of-Information Minimization in a Remote Monitoring System",
IEEE Information Theory Workshop (IEEE ITW) (M) 2024
29. [VTC](#) Aiwei Lei, Di Zhang, **Aimin Li**, and Evgeny Khorov. "Closed-Form Capacity Analysis of Weibull Fading Channels for Satellite-Ground Communications",
IEEE Vehicular Technology Conference (IEEE VTC) (M) 2026
30. [IAC](#) Sajjad Baghaee, **Aimin Li**, Shanmugarajan Balasubramanian, and Elif Uysal*. "Goal-Oriented Bundle Management and Flow Control for Deep-Space Communications via Reinforcement Learning",
77th International Astronautical Congress (IAC) 2026
31. [ICCT](#) Mingxiao Sun, Di Zhang, Shaobo Jia, and **Aimin Li**. "Packet Management of AoI in the Finite Block-length Regime",
IEEE International Conference on Communication Technology (IEEE ICCT) (M) 2024
32. [VTC](#) Jiaming Zhang, Shaohua Wu*, **Aimin Li**, Jian Jiao, and Qinyu Zhang. "Analysis of GEO Satellite Relay Coded Systems",
IEEE Vehicular Technology Conference (VTC2022-Fall) 2022
33. [INFOCOM](#) Kunpeng Liu, **Aimin Li**, and Shaohua Wu*. "Deep Reinforcement Learning-Assisted Age-optimal Transmission Policy for HARQ-aided NOMA Networks",
IEEE INFOCOM Workshops (M) 2023
34. [ICC](#) Siqi Meng, Shaohua Wu*, **Aimin Li**, Jian Jiao, Ning Zhang, and Qinyu Zhang. "Analysis and Optimization of Spinal Codes over the BSC: from the AoI Perspective",
IEEE ICC Workshops (M) 2021

SELECTED MANUSCRIPTS UNDER REVIEW

A. First-Author Submissions

Journal Articles

35. [TMC](#) Haojie Bai[#], **Aimin Li**[#], Ruoyu Yao, Xiongwei Zhao, Tingting Zhang, Xing Zhang, Caixiong Li, Lin Gao, and Jun Ma. "SCORP: Scene-Consistent Multi-agent Diffusion Planning with Stable Online Reinforcement Post-Training for Cooperative Driving",
submitted to IEEE Trans. Mob. Comput. (Under Review) 2026

Conference Proceedings

36. [IROS](#) Haojie Bai[#], **Aimin Li**[#], Ruoyu Yao, Xiongwei Zhao, Tingting Zhang, Xing Zhang, Caixiong Li, Lin Gao, and Jun Ma. "Multi-ORFT: Stable Online Reinforcement Fine-Tuning for Multi-Agent Diffusion Planning in Cooperative Driving",
submitted to IEEE/RSJ International Conference on Intelligent Robots and Systems (IEEE IROS). (Under Review) 2026
37. [ASILOMAR](#) Sayam Chakraborty[#], **Aimin Li**[#], Yigit Ince, Sajjad Baghaee, and Elif Uysal*. "ASTRA: A Mean-Field Approach with Asynchronous Capture-SIC",
submitted to Asilomar Conference on Signals, Systems, and Computers 2026. (Under Review) (M) 2026

B. Collaborative Submissions

Journal Articles

38. [TIT](#) Lulu Song, Di Zhang, **Aimin Li**, and Tingting Zhang. "Source PAC Coding for Low-Latency Secret Key Generation in Short-Blocklength Regime",
submitted to IEEE Trans. Inf. Theory. (Under Review) (M) 2025
39. [TMC](#) Adem Utku Atasayar, **Aimin Li**[†], Cagri Ari, and Elif Uysal*. "Age Optimal Sampling and Routing under Intermittent Links and Energy Constraints",
submitted to IEEE Trans. Mob. Comput. (Under Review) (M) 2026
40. [TWC](#) Guoquan Chen, Yajing Deng, **Aimin Li**, and Shaohua Wu. "Permanent Connection Guaranteed Resilient LEO Constellation for Remote Area Communications",
submitted to IEEE Trans. Wirel. Commun. (Under Review) 2025
41. [CN](#) Shanmugarajan Balasubramanian, Sajjad Baghaee, **Aimin Li**, and Elif Uysal*. "PPO-Based AoI and Energy Aware Scheduling in Delay Tolerant Networks",
submitted to Computer Networks. (Under Review) (M) 2026

Conference Proceedings

42. [MobiHoc](#) Adem Utku Atasayar, **Aimin Li**, and Elif Uysal*. "When Worse Routes Become Better: Preemptive Multi-Route Optimization for Age of Information",
submitted to ACM MobiHoc 2026. (Under Review) (M) 2026
43. [GLOBECOM](#) Yinglei Yang, Di Zhang, **Aimin Li**, Yifei Shen, Alexey Frolov, and Bo Gu. "On the Multi-Weight Shift-Sum Iterative Decoding of Short BCH Codes",
submitted to IEEE Global Communications Conference (IEEE GLOBECOM). (Under Review) (M) 2026

44. **SIGCOMM** Ayşe Nur Pehlivanoglu, **Aimin Li**, and Elif Uysal*. “In-Orbit Intelligence or Ground Offloading? Inference Freshness under Intermittent Satellite Connectivity”, submitted to *ACM SIGCOMM Workshop (LEO-NET)* 2026. (Under Review) (M) 2026
45. **ITW** Burak Karasakal, **Aimin Li**, and Elif Uysal*. “Age of Information in Time-Varying Multi-Priority Queues”, submitted to *IEEE Information Theory Workshop (IEEE ITW)* 2026. (Under Review) (M) 2026

RESEARCH LEADERSHIP AND MENTORSHIP

Research Leadership Initiated and led 14 student-involved research collaborations across B.S., M.S., and Ph.D. stages. I formulated the research questions, mathematical models, and solution frameworks; junior collaborators carried out analysis, simulations, and implementation under my supervision. Outcomes include publications/submissions in IEEE TMC, TWC, IOTJ, ISIT, MobiHoc, ICC, ITW, GLOBECOM, IAC, Asilomar, and student placements at KTH, Caltech, HKU, Huawei, HIT, METU, Zhengzhou University, NUS, and ISRO.

Student mentorship and collaboration

Student	Mentorship / outcome
Adem Utku Atasayar	2025–2026; co-advised M.S. thesis; MobiHoc/TMC; Ph.D. at METU.
Xiaomeng Chen	2023–2026; M.S. collaboration; TWC/IOTJ/ISIT/ITW; admitted to KTH Ph.D.
Kunpeng Liu	2024–2026; M.S. collaboration; ICC/INFOCOM Workshop; now at Huawei.
Siqi Meng	2020–2024; B.S. and Ph.D. collaboration; MWC/TCOM/TVT/JSAT; Ph.D. at HIT.
Burak Karasakal	2025–2026; B.S. collaboration; ITW; admitted to Caltech Ph.D.
Aiwei Lei	2026; B.S. collaboration; VTC; admitted to HKU Ph.D.
Haoxu Wu	2025–2026; Ph.D. collaboration; accepted IEEE TMC paper.
Lulu Song	2026; Ph.D. collaboration; now Ph.D. at Zhengzhou University.
Yigit Ince	2025–2026; M.S. collaboration; ISIT; METU.
Yinglei Yang	2026; M.S. collaboration at Zhengzhou University; GLOBECOM.
Ayşe Nur Pehlivanoglu	2025–2026; B.S. and M.S. collaboration; SIGCOMM Workshop; METU.
Shanmugarajan Anand	2026; B.S. collaboration; IAC; admitted to NUS M.S. program.
Sayam Chakraborty	2026; B.S. collaboration; Asilomar; works at Indian Space Research Organisation (ISRO).

AWARDS

Award	Distinction	Year
Best Dissertation Award Nomination	One of only 2 nominees in the School; only EE Department nominee.	2025
National Scholarship of China (twice)	China’s highest-level student scholarship; top 0.2% nationwide.	2020, 2024
Provincial Outstanding Graduate Award	Top 0.4% in Heilongjiang Province.	2025
Best Undergraduate Thesis Award	Only 2 recipients in the EE Department.	2020
5th Place, IEEE GLOBECOM 4MT Thesis Competition	Ranked 5th among 15 global finalists worldwide.	2024
IEEE GLOBECOM Travel Grant	Top 15 Ph.D. students worldwide.	2024
Candidate for NSFC Youth Project	One of fewer than 600 Ph.D. students in China.	2024
HIT “Dianzi” Research Student Program	Independent PI project; only 1 Ph.D. student in the EE Department.	2023
HIT Special Merit Doctoral Scholarship	University-President award; only 8 Ph.D. recipients university-wide.	2020
Award for HIT Outstanding Graduate Students	Top 10% at Harbin Institute of Technology.	2025
Outstanding Participant, Tsinghua University Summer Camp	Qualified for postgraduate recommendation to Tsinghua University.	2019
HIT Outstanding Student Award (twice)	Top 10% at Harbin Institute of Technology.	2017, 2019
Gongjin Scholarship for Overall Excellence	Only 3 recipients in the EE Department.	2018
Provincial Merit Student	Top 1% in Heilongjiang Province.	2018
First Prize, Chinese College Student Math Competition (twice)	National-level mathematics competition.	2016, 2017
First-Class Scholarship for Comprehensive Excellence	Top 5% at Harbin Institute of Technology.	2017

TEACHING EXPERIENCE

Communication Theory (Undergraduate Course)	Teaching Assistant with Prof. Shaohua Wu, HITSZ.	2020
Foundations of Information Theory (Graduate Course)	Teaching Assistant with Prof. Shaohua Wu, HITSZ.	2022

ACADEMIC SERVICE

Conference Session Chair IEEE ITW 2024 Age of Information Session; IEEE GLOBECOM 2024 Distributed Learning Session.

Reviewer

Journals: IEEE Trans. Inf. Theory (TIT); IEEE J. Sel. Areas Commun. (JSAC); IEEE Trans. Mob. Comput. (TMC); IEEE Trans. Commun. (TCOM); IEEE Trans. Wirel. Commun. (TWC); IEEE Trans. Neural Netw. Learn. Syst. (TNNLS); IEEE Trans. Veh. Technol. (TVT); IEEE Trans. Cogn. Commun. Netw. (TCCN); IEEE Commun. Lett. (CL); IEEE Wirel. Commun. Lett. (WCL); Elsevier Computer Networks.

Conference reviewer

IEEE ISIT	IEEE INFOCOM	IEEE ICC
IEEE GLOBECOM	IEEE WCNC	IEEE VTC
IWCMC		

Volunteer IEEE 6G Summit 2023, Singapore; IEEE VTC-Spring 2024, Singapore; IEEE 6G Summit 2024, Singapore.

REFERENCES

Elif Uysal, IEEE Fellow, Postdoctoral Advisor, METU, uelif@metu.edu.tr; Shaohua Wu, Ph.D. Advisor, HITSZ, hitwush@hit.edu.cn; Sumei Sun, IEEE Fellow, Ph.D. Co-Advisor, A*STAR, sunsm@i2r.a-star.edu.sg; Di Zhang, Professor, Sun Yat-sen University, dr.di.zhang@ieee.org.